
CODE OF BOOTH MULTIPLIER

/home/dhruv/cao_lab/vivado_proj/cao_lab_u24ev057/rtl/booth/booth_mul.vhd [VHDL]

```
1  -----
2  --
3  -- Company:
4  -- Engineer:
5  --
6  -- Create Date: 09/10/2026 04:26:32 PM
7  -- Design Name:
8  -- Module Name: booth_mul - Behavioral
9  -- Project Name:
10 -- Target Devices:
11 -- Tool Versions:
12 -- Description:
13 --
14 -- Dependencies:
15 --
16 -- Revision:
17 -- Revision 0.01 - File Created
18 -- Additional Comments:
19 --
20 -----
21 --
22 library IEEE;
23 use IEEE.STD_LOGIC_1164.ALL;
24
25 -- Uncomment the following library declaration if using
26 -- arithmetic functions with Signed or Unsigned values
27 use IEEE.NUMERIC_STD.ALL;
28 --library UNISIM;
29 --use UNISIM.VComponents.all;
30
31 entity booth_mul is
32     Port ( a : in signed (3 downto 0);
33           b : in signed (3 downto 0);
34           p : out signed (7 downto 0));
35 end booth_mul;
36
37 architecture Behavioral of booth_mul is
38 begin
39
40     process (a , b)
41         variable q_pair : std_logic_vector(1 downto 0);
42         variable acc : signed (3 downto 0);
43         variable q : signed (3 downto 0);
44         variable total : std_logic_vector (8 downto 0);
45     end process;
```

46	variable w : integer := 3;
47	
48	begin
49	q_pair := "00";
50	acc := "0000";
51	q := b;
52	total := std_logic_vector(acc) & std_logic_vector(q) & q_pair(0);
53	
54	for i in 0 to w loop
55	
56	q_pair := total(1 downto 0);
57	
58	case(q_pair) is
59	when "01" => acc := acc + a;
60	when "10" => acc := acc -a ;
61	when others => null ;
62	end case;
63	
64	total := std_logic_vector(acc) & std_logic_vector(q) & q_pair(0);
65	total := total(8) & total(8 downto 1);
66	acc := signed(total(8 downto 5));
67	q := signed(total(4 downto 1));
68	
69	end loop;
70	
71	p <= signed(total(8 downto 1));
72	
73	end process;
74	
75	end Behavioral;

CODE OF TEST BENCH FOR BOOTH MULTIPLIER

/home/dhruv/cao_lab/vivado_proj/cao_lab_u24ev057/tb/booth/booth_mul_tb.vhd [VHDL]	
1	-----
	--
2	-- Company:
3	-- Engineer:
4	--
5	-- Create Date: 09/10/2026 04:26:32 PM
6	-- Design Name:
7	-- Module Name: booth_mul_tb - Behavioral
8	-- Project Name:
9	-- Target Devices:
10	-- Tool Versions:
11	-- Description:
12	--
13	-- Dependencies:
14	--
15	-- Revision:

```

16 -- Revision 0.01 - File Created
17 -- Additional Comments:
18 --
19 -----
20 --
21
22 library IEEE;
23 use IEEE.STD_LOGIC_1164.ALL;
24
25 -- Uncomment the following library declaration if using
26 -- arithmetic functions with Signed or Unsigned values
27 use IEEE.NUMERIC_STD.ALL;
28
29 -- Uncomment the following library declaration if instantiating
30 -- any Xilinx leaf cells in this code.
31 --library UNISIM;
32 --use UNISIM.VComponents.all;
33
34 entity booth_mul_tb is
35 -- Port ( );
36 end booth_mul_tb;
37
38 architecture testing of booth_mul_tb is
39
40
41     component booth_mul is
42         Port ( a : in signed (3 downto 0);
43               b : in signed (3 downto 0);
44               p : out signed (7 downto 0));
45     end component;
46
47
48     signal a : signed(3 downto 0);
49     signal b : signed(3 downto 0);
50     signal p : signed(7 downto 0);
51
52 begin
53
54
55     booth_multiplier_test : booth_mul port map(
56         a => a,
57         b => b,
58         p => p
59     );
60
61
62     process
63
64     begin

```

65	a <= "0101";
66	b <= "1010";
67	
68	wait for 10ns;
69	
70	a <= "0011";
71	b <= "1100";
72	
73	wait for 10ns;
74	
75	a <= "1010";
76	b <= "1100";
77	
78	wait for 10ns;
79	
80	a <= "0001";
81	b <= "1000";
82	
83	wait for 10ns;
84	
85	a <= "1110";
86	b <= "1011";
87	
88	wait for 10ns;
89	
90	a <= "1111";
91	b <= "1111";
92	
93	wait for 10ns;
94	
95	a <= "0000";
96	b <= "0000";
97	
98	wait for 10ns;
99	
100	a <= "1010";
101	b <= "1111";
102	
103	wait for 10ns;
104	
105	a <= "0001";
106	b <= "1111";
107	
108	wait for 10ns;
109	
110	a <= "1010";
111	b <= "1111";
112	
113	
114	wait;

115	
116	end process;
117	
118	
119	
120	end testing;

SCREENSHOT OF CODE

The screenshot displays the Vivado IDE interface. The left sidebar shows the 'PROJECT MANAGER' with a tree view containing 'Settings', 'Add Sources', 'Language Templates', 'IP Catalog', 'IP INTEGRATOR', 'SIMULATION', 'RTL ANALYSIS', 'SYNTHESIS', 'IMPLEMENTATION', and 'PROGRAM AND DEBUG'. The main editor window shows the 'booth_mul.vhd' file. The code is as follows:

```
25 -- Uncomment the following library declaration if using
26 -- arithmetic functions with Signed or Unsigned values
27 use IEEE.NUMERIC_STD.ALL;
28 --library UNISIM;
29 --use UNISIM.VComponents.all;
30
31 entity booth_mul is
32   Port ( a : in signed (3 downto 0);
33         b : in signed (3 downto 0);
34         p : out signed (7 downto 0));
35 end booth_mul;
36
37 architecture Behavioral of booth_mul is
38 begin
39
40   process (a, b)
41     variable q_pair : std_logic_vector(1 downto 0);
42     variable acc : signed (3 downto 0);
43     variable q : signed (3 downto 0);
44     variable total : std_logic_vector (8 downto 0);
45     variable w : integer := 3;
46
47   begin
48     q_pair := "00";
49     acc := "0000";
50     q := b;
51     total := std_logic_vector(acc) & std_logic_vector(q) & q_pair(0);
52
53     for i in 0 to w loop
54
55       q_pair := total(1 downto 0);
56
57       case(q_pair) is
58         when "00" => acc := acc + a;
59         when "01" => acc := acc - a;
60         when others => null;
61       end case;
62
63       total := std_logic_vector(acc) & std_logic_vector(q) & q_pair(0);
64       total := total(8) & total(8 downto 1);
65       acc := signed(total(8 downto 5));
66       q := signed(total(4 downto 1));
67
68     end loop;
69
70     p <= signed(total(8 downto 1));
71
72   end process;
73 end Behavioral;
```

The status bar at the bottom indicates '36:1 Insert VHDL'.

Figure 1: Design

The screenshot displays the Vivado IDE interface. The left sidebar shows the 'PROJECT MANAGER' with a tree view containing 'Settings', 'Add Sources', 'Language Templates', 'IP Catalog', 'IP INTEGRATOR', 'SIMULATION', 'RTL ANALYSIS', 'SYNTHESIS', 'IMPLEMENTATION', and 'PROGRAM AND DEBUG'. The main editor window shows the 'booth_mul_tb.vhd' file. The code is as follows:

```
31 --library UNISIM;
32 --use UNISIM.VComponents.all;
33
34 entity booth_mul_tb is
35   Port ( );
36 end booth_mul_tb;
37
38 architecture testing of booth_mul_tb is
39
40
41   component booth_mul is
42     Port ( a : in signed (3 downto 0);
43           b : in signed (3 downto 0);
44           p : out signed (7 downto 0));
45   end component;
46
47   signal a : signed(3 downto 0);
48   signal b : signed(3 downto 0);
49   signal p : signed(7 downto 0);
50
51 begin
52
53   booth_multiplier_test : booth_mul port map(
54     a => a,
55     b => b,
56     p => p
57   );
58
59   process
60   begin
61     a <= "0101";
62     b <= "1010";
63
64     wait for 10ns;
65
66     a <= "0011";
67     b <= "1100";
68
69     wait for 10ns;
70
71     a <= "1010";
72     b <= "1100";
73
74     wait for 10ns;
75
76     a <= "0001";
77     b <= "1000";
78
79     wait for 10ns;
80
81     a <= "0001";
82     b <= "1000";
83
84   end process;
```

The status bar at the bottom indicates '38:1 Insert VHDL'.

Figure 2: Tb

WAVEFORM

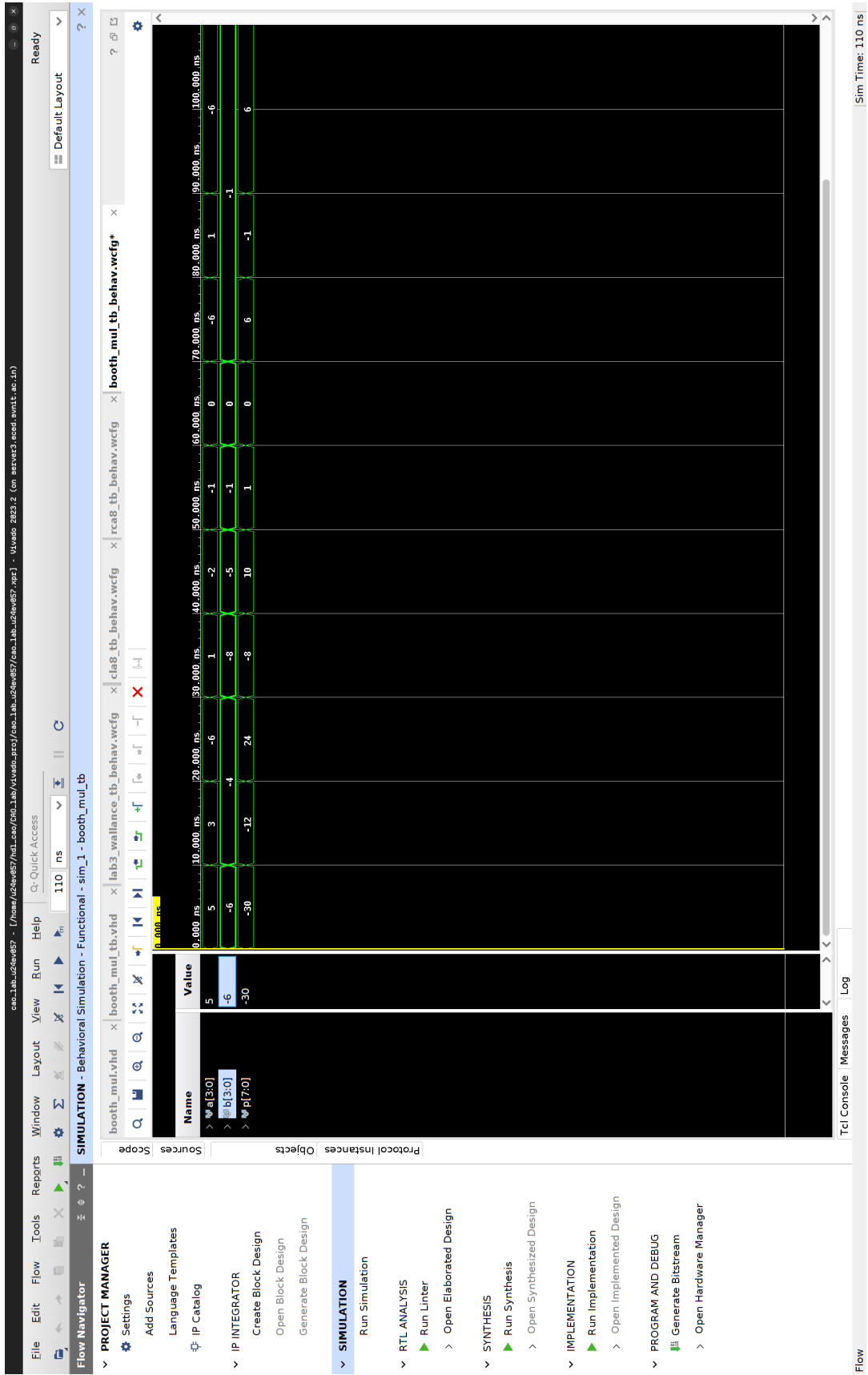


Figure 3: Waveform

